

PRACTICAL WEEK-1(OSSP)

NAME: VARSHITH COMPLE

```
Last login: Mon Aug 17 10:23:24 on console
/Users/varshithcomple/.zprofile:1: no such file or directory: /usr/local/bin/brew
varshithcomple@VARSHITHs-MacBook-Air ~ % ls
[Applications    Downloads      Library       Pictures       task1.c
Desktop          first         Movies        Public         task1.c.save
Documents        first.c       Music         task1
varshithcomple@VARSHITHs-MacBook-Air ~ % pwd
/Users/varshithcomple

svlang: SystemVerilog compiler and language services
systemd-isp: language server for systemd unit files
tracy-genomics: Basecalling, alignment, assembly and deconvolution of Sanger chromatograms
ubntree: Live USB device tree in your terminal
vapoursynth-wraps: VapourSynth Zip Image Process
vi-sql: Terminal UI for SQL databases
xmedcon: Medical image conversion toolkit
  => New Casks
android-cil: Command-line interface for Android app development with AI agents
bb: IDE for running and orchestrating coding agents
canario: Terminal emulator
casahub: Native GUI for Homebrew casks
diversion: Cloud-native version control CLI and agent
diversion-app: Desktop app for Diversion version control
dsk-control: Menu bar pane for DeepSeek V4 via DwarfStar
easydm: One click DMG installs
esphome-device-builder: Desktop app to create, edit and install your ESPHome device configurations
font-jetstandard
font-mexon-bazzi
font-mexon-football-gothic
font-mexon-kart-gothic
font-mexon-lvl-gothic
font-mexon-lv2-gothic
font-mexon-maplestory
greenmakerlts: Complete development tool for making 2D games
gleam: workspace search and AI assistant
grok-bot: AI teammates that work across your apps and tools
headroom: Reduce token usage for Claude Code and CodeX
kilocrew: Persistent AI development workspace with multi-agent support
koreader: Document viewer for e-ink devices
meetingrecorder: Recorder for meetings capturing mic and system audio
mongrel: Database workbench with terminals, containers, Kubernetes, and API client
muse-code: Interactive terminal coding agent
owid: On-device OCR for screenshots, images, and PDFs
petdex: Desktop pet that reflects coding agent activity
sentry-cli: Command-line utility to interact with Sentry
sina-finance: Stock market data and financial news platform
space-rabbit: Removes animations when switching between Spaces
subtitle-edit: Subtitle editor
tauritevrn: SillyTavern-compatible native client
tesseract: Tesseract that highlights scripts in real time as you speak
warp-agent-cil: Agentic development environment for command-line workflows
zcode: AI-assisted development environment

You have 13 outdated formulae installed.

Warning: No available formula with the name "clang". Did you mean clang1, c1n, c1ng, c1ean, v1ang, f1ang, c1ingo or c1ac?
  -- Searching for similarly named formulae and casks...
  -- Formulae
clang-build-analyzer      clang-format@11 (deprecated)  clang-uml                  emacs-clang-complete-async  c1ng      clang      v1ang      c1ingo
clang-format              clang-include-graph          clang1                      c1n                        c1ean
To install clang-build-analyzer, run:
brew install clang-build-analyzer

  -- Casks
xcodclangformat (deprecated)
To install xcodclangformat (deprecated), run:
brew install --cask xcodclangformat (deprecated)
varshithcomple@VARSHITHs-MacBook-Air ~ % brew install clang-build-analyzer
  --> Downloading bottle manifests
  --> Bottle Manifest clang-build-analyzer (1.6.0)
  --> Would install 1 formula:
clang-build-analyzer
  --> Fetching downloads for: clang-build-analyzer
  --> Bottle clang-build-analyzer (1.6.0)
  --> Pouring clang-build-analyzer--1.6.0.arm64.tahoe.bottle.tar.gz
  --> Downloaded 194.1KB/194.1KB
  --> Running 'brew cleanup clang-build-analyzer'...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
varshithcomple@VARSHITHs-MacBook-Air ~ % gcc notes.c -o notes
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
varshithcomple@VARSHITHs-MacBook-Air ~ %
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clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
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clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if (pid == 0)
    {
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else
    {
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
varshithcomple@VARSHITHs-MacBook-Air ~ % gcc -mmacosx-version-min=26.0 notes.c -o notes
varshithcomple@VARSHITHs-MacBook-Air ~ % gcc notes.c -onotes
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
varshithcomple@VARSHITHs-MacBook-Air ~ % ./notes
This is the Parent Process.
Parent PID = 49737
Child PID = 49738
This is the Child Process.
Child PID = 49738
varshithcomple@VARSHITHs-MacBook-Air ~ % nano notes.c
varshithcomple@VARSHITHs-MacBook-Air ~ %
```

```
varshithcomple@VARSHITHs-MacBook-Air ~ % whoami
varshithcomple
varshithcomple@VARSHITHs-MacBook-Air ~ % who
varshithcomple console 17 Aug 10:23
varshithcomple ttyne000 19 Aug 07:22
varshithcomple@VARSHITHs-MacBook-Air ~ % id
uid=0(varshithcomple) gid=20(staff) groups=20(staff),101(access_bpf),12(everyone),51(localaccounts),791(appserverusr),88(admin),81(appserveradm),701(com.apple.sharepoint.group.1),331(appstore),90(_loadadmin),100(_operator),204(_developer),209(_analyticsusers),390(com.apple.access.tftp),398(com.apple.access.screensharing),399(com.apple.access.ssh),400(com.apple.access.remote_w)
varshithcomple@VARSHITHs-MacBook-Air ~ % hostname
VARSHITHs-MacBook-Air.local
varshithcomple@VARSHITHs-MacBook-Air ~ % date
Wed Aug 19 07:29:17 IST 2020
varshithcomple@VARSHITHs-MacBook-Air ~ % cal
      August 2020
Su Mo Tu We Th Fr Sa
1
2 3 4 5 6 7 8
9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
varshithcomple@VARSHITHs-MacBook-Air ~ % uname
Darwin
varshithcomple@VARSHITHs-MacBook-Air ~ % uptime
7:38 up 1 day, 21:14, 2 users, load averages: 1.54 1.43 1.35
varshithcomple@VARSHITHs-MacBook-Air ~ % df -h
Filesystem      Size   Used  Avail Capacity  iused ifree %iused  Mounted on
/dev/disk3s1s1  4680G  120G   390G    3%    49K  3.6G    0%    /
devfs            345K   345K    0%   1.2k    0 100%    /dev
/dev/disk3s6    4680G  1.8G   339G    4%    1.3G  3.6G    0%    /System/Volumes/VW
/dev/disk3s2    4680G  8.5G   339G    1%   1.5k  3.6G    0%    /System/Volumes/Preboot
/dev/disk3s4    4680G  7.4M   339G    1%   119  3.6G    0%    /System/Volumes/Update
/dev/disk3s2    5880G  6.0M   482M    0%    1 4.9M    0%    /System/Volumes/errts
/dev/disk3s1    5880G  6.0M   482M    0%   32 4.9M    0%    /System/Volumes/ISOreboot
/dev/disk3s3    5880G  1.5M   482M    1%   98 4.9M    0%    /System/Volumes/hardware
/dev/disk3s8    4680G  90G   339G    2%   85k  3.6G    0%    /System/Volumes/data
map auto_home   0B     0B    0B   100%    0 0      -    /System/Volumes/Data/home
```

[illegible]

UW PICO 5.09

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command execution failed.\n");
        exit(1);
    }
    else
    {
        wait(NULL);

        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }

    return 0;
}
```

```
varshithcomple@VARSHITHs-MacBook-Air ~ % nano exec.c
varshithcomple@VARSHITHs-MacBook-Air ~ % gcc exec.c -oexec
clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]
varshithcomple@VARSHITHs-MacBook-Air ~ % ./exec
Enter a Linux command: ls
```

```
Child Process
Child PID : 53578
Applications  Documents  exec      first      Library    Music      notes.c    Pictures   task1      task1.c.save
Desktop      Downloads  exec.c    first.c    Movies
Parent Process
Parent PID : 53477
Child process completed.
```

```
[varshithcomple@VARSHITHs-MacBook-Air ~ % uname
Darwin
[varshithcomple@VARSHITHs-MacBook-Air ~ % lscpu
zsh: command not found: lscpu
[varshithcomple@VARSHITHs-MacBook-Air ~ % lsblk
zsh: command not found: lsblk
[varshithcomple@VARSHITHs-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 39679 ttys000    0:00.13 -zsh
[varshithcomple@VARSHITHs-MacBook-Air ~ % top
varshithcomple@VARSHITHs-MacBook-Air ~ % >
```